(Please write your Enrollment Number) Enrollment No. 0480 117 7022

MID-TERM EXAMINATION

Course Name: B. Tech (AI & DS) Semester:6

(March 2025) OFF LINE mode

Subject Code: BAI 302	Subject: Natural Language Processing
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

		(2.5*4=10)	CO Mapping
Q1			
	a) Explain how **tokenization** works in NLP with an example.	Unit 2	Co2
	b) What is the role of a **lexicon** in language processing?	Unit 1	Co1
	c) What is the formula for **TF-IDF**?	Unit 2	Co1
	d) How does syntactic parsing help in understanding sentence structure?	Unit 1	Co2
	UNIT I		**CO Mapping**
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) Differentiate between **phonology, morphology, syntax, and semantics.**	5	Co1
	b) Identify **syntactic ambiguity** in the sentence **"He saw the girl with the telescope."** What are the **limitations** of syntactic parsing in real-world applications?	5	Co3,4
	c) List and explain the different types of **ambiguity** in NLP with examples	5	Co1,2
	UNIT II		**CO Mapping**
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) Explain how **JSON and XML** are used for storing and exchanging text data. What are their key differences?	5	Co1,2
	b) Using **BoW**, represent the sentences: "NLP is great" "NLP is fun" **Vocabulary:** {NLP, is, great, fun}	5	Co3
	c) Define a **text corpus** in NLP and list three different types of corpora used in text processing.	5	Co1